

# Formal Certification of Neural Lyapunov Functions for Transient Stability in Lossy Power Networks

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**Abstract**—A neural Lyapunov function trained against randomly sampled violations cannot be certified, because a certificate quantifies over every point in a region while sampling only ever visits a measure-zero subset, so a network can report a 99.98% empirical satisfaction rate and still fail the Lyapunov conditions on a set that random sampling never lands on. We reproduce the Lyapunov-regularized reinforcement learning of Cui and Zhang for the lossy Kron-reduced New England 39-bus swing model, then show with the neural network verifier CROWN that the as-trained value function violates both the positive-definiteness condition and the Lie-derivative condition through genuine counterexamples that a directed search finds immediately while 200,000 random samples find none. We then close the loop, using the verifier itself as the sampler and feeding its counterexamples back into training, so the same architecture that certifies over nothing becomes certifiable over a real region: on the projected gen-5 slice, a two-dimensional projection that holds the network’s other buses at equilibrium and matches the authors’ Fig. 2, and not on the full state, counterexample-guided training certifies the Lie-derivative condition out to a sublevel radius  $\rho = 2.0$  and an exponential decrease rate  $\beta$  up to 3.97, both audited by an independent attack and cross-checked against a second verifier. The certified region collapses as the state dimension grows, which is quantified rather than hidden and which points directly at where the next work has to go.

**Index Terms**—Neural Lyapunov functions, transient stability, formal verification, CROWN, branch-and-bound, counterexample-guided training.

## I. INTRODUCTION

Transient stability of a power network asks whether the generators, after a disturbance, swing back to a synchronous equilibrium rather than losing synchronism. The classical certificate is an energy function, but for a network with resistive losses the conductance coupling enters the swing equations as a term  $G_{ij} \cos(\delta_i - \delta_j)$  that is not the gradient of any scalar, so no analytic energy function exists [1]. That single term is why a valid Lyapunov function for lossy frequency dynamics has been an open problem for decades, and it is the entire reason the frequency-control problem is interesting while the voltage-control problem, whose dynamics are linear, is not.

Cui and Zhang sidestep the non-existence result by *learning* a Lyapunov function  $V_\phi$  as a neural network, jointly with a decentralized controller, under a training loss that penalizes violations of the Lyapunov conditions on sampled states [2]. Their method works well empirically, and the last sentence of their paper names the gap it leaves open: an important direction is to verify whether the learned function satisfies the Lyapunov conditions for *all* points in a region, not just

on samples. They report the Lie derivative is slightly positive on about 0.1% of sampled points after training. Sampling is not a proof, because the conditions are universally quantified over a continuous region while any sample set is finite, so the question their paper poses is exactly the question a verifier answers.

This paper answers it. We reproduce their training, we use the neural network verifier CROWN [3] with branch-and-bound over the nonlinear swing dynamics to certify the Lyapunov conditions over a region rather than a sample, and where the as-trained network fails we close the loop with counterexample-guided training. The contribution is not a single verified box, it is the demonstration that the sampling-to-certification gap is real and directed, that it is the same gap in the condition that was supposed to be trivial, and that feeding a sound verifier’s counterexamples back into training is what converts an uncertifiable network into a certifiable one over a real region.

## II. RELATED WORK

Certifying a learned Lyapunov function means proving a nonlinear inequality over a continuous region, and the tools differ in what nonlinearities they admit and how they scale. Satisfiability-modulo-theories methods with a  $\delta$ -complete solver certify neural Lyapunov controllers exactly [4], which is the approach Cui and Zhang cite, but they scale poorly and are typically tractable only for small systems, which is why their own paper notes the limitation. Mixed-integer encodings require piecewise-linear activations and so cannot represent the ELU network used here at all. Sum-of-squares programming requires polynomial dynamics and so cannot represent the sin and cos of the swing equations. Learned-dynamics constructions that build stability in by design [5] avoid verification but change the model class.

Bound propagation with branch-and-bound occupies a different point. CROWN propagates a sound linear relaxation of a general nonlinear computation graph [3], its automatic-differentiation implementation auto\_LiRPA bounds arbitrary graphs [6],  $\beta$ -CROWN adds per-neuron branching for completeness [7], and GenBaB extends branch-and-bound to general nonlinearities including sin, cos, and the ELU derivative [8]. The bounds are GPU-parallel across subdomains and, being differentiable, can be trained against [9]. This is the reason CROWN, and not SMT, is the right tool for a lossy swing model with an ELU value function: it handles the exact operators the problem contains, and it hands back a counterexample when it fails rather than only a verdict. Our

formulation of the Lie derivative as a boundable graph and the sublevel-set region-of-attraction follow the neural control verification line [10] and its tutorial [11].

### III. PROBLEM FORMULATION

#### A. Lossy swing dynamics

The New England 39-bus system is Kron-reduced to its  $n = 10$  generator buses. With state  $x = (\delta, \omega) \in \mathbb{R}^{2n}$  the swing dynamics are [2]

$$\dot{\delta}_i = \omega_i, \quad (1)$$

$$M_i \dot{\omega}_i = P_i - D_i \omega_i - u_i(\omega_i) - \sum_j B_{ij} \sin(\delta_i - \delta_j) - \sum_j G_{ij} \cos(\delta_i - \delta_j) \quad (2)$$

where  $B_{ij}$  is the susceptance coupling and  $G_{ij}$  is the conductance coupling produced by network losses. We match the authors' training dynamics, which integrate  $\dot{\delta}_i = \omega_{\text{scale}} \omega_i$  with  $\omega_{\text{scale}} = 2\pi$  because their  $\omega$  is a per-unit frequency in Hz, and which fold the damping  $D_i \omega_i$  into the controller, so the certified closed loop is  $f(x, u(x))$  with  $u$  supplying the damping.

#### B. Controller and Lyapunov network

The certified controller is the saturated linear droop law  $u_i(\omega_i) = \text{sat}(c_i \omega_i)$  that the authors use while training  $V_\phi$ ; it is piecewise linear, so CROWN handles it exactly. The value function  $V_\phi(\delta, \omega)$  is a single-hidden-layer network with ELU activations, chosen for a smooth bounded derivative, evaluated on the reference-subtracted coordinates  $z = [\delta_i - \delta_1, \omega_i - \omega_1]$  of dimension  $2(n-1) = 18$ , which removes the rotational symmetry of the swing model so the equilibrium is an isolated point.

#### C. Lyapunov conditions

With equilibrium  $x^*$  and  $V^* = V(x^*)$ , the conditions are [2]

$$(4a) \quad V(x) > V^* \quad \text{for } x \neq x^*, \quad (3)$$

$$(4b) \quad \dot{V}(x) = \nabla V(x) \cdot f(x, u(x)) < 0 \quad \text{for } x \neq x^*, \quad (4)$$

$$(4c) \quad \dot{V}(x^*) = 0, \quad (5)$$

and Proposition 2 strengthens (4b) to an exponential rate,  $\dot{V}(x) \leq -\beta(V(x) - V^*)$ . She named (4a) for certification. We certify it because she asked, but (4a) is a property of  $V$  alone and never touches the dynamics, so the condition that actually needs the nonlinear swing equations is (4b) and its Proposition-2 form, and that is where the contribution is.

### IV. VERIFICATION FORMULATION

#### A. The Lie derivative as a boundable graph

Condition (4b) requires bounding  $\dot{V}(x) = \nabla V(x) \cdot f(x, u(x))$  over a box, which means the gradient of  $V$  has to be part of the graph the verifier bounds. For the single-hidden-layer ELU network the gradient is analytic,

$$\nabla_z V = (W_2 \odot \text{ELU}'(W_1 z + b_1)) W_1, \quad \text{ELU}'(a) = e^{-\text{relu}(-a)}, \quad (6)$$

and every operation in it, together with the sin and cos of  $f$ , is a natively boundable operator, so we express the whole scalar  $F(x) = -\dot{V}(x)$  as one computation graph and bound it with CROWN directly. This specializes the general-purpose route, in which auto\_LiRPA's Jacobian operator expands gradient nodes and GenBaB branches the sin, cos, and ELU nonlinearities [8], [11]; for a one-hidden-layer network the analytic gradient graph is exact and its bounds are tighter, so we use it and keep the Jacobian route available for deeper networks. We also replace the ELU by the identity  $\text{ELU}(x) = \text{relu}(x) - \text{relu}(1 - e^{-\text{relu}(-x)})$ , which matches `torch.nn.ELU` to  $6 \times 10^{-8}$  and is boundable, since this build of auto\_LiRPA has no native ELU bound.

#### B. Certification and the equilibrium

We certify  $F(x) \geq -\epsilon$  over a box  $B$  by propagating a sound CROWN lower bound and branching  $B$  on its widest dimension where the bound is inconclusive; a nonnegative bound on every leaf verifies the box, and a leaf that cannot be split hands back either a genuine counterexample with  $F < -\epsilon$  (the property is false) or an unresolved region (the bound stayed loose). These two outcomes are never conflated. Every verified box is then re-checked by an independent multi-start projected-gradient attack, and a certificate is rejected if the attack finds any  $F < -\epsilon$ , which guards against an unsound bound.

The Lie derivative is zero at the equilibrium by construction, so  $F = -\dot{V}$  is zero there and no region containing  $x^*$  certifies (4b) as a strict inequality. We handle this the standard way, certifying on an annulus that excludes a small ball around  $x^*$ , and reporting an explicit tolerance  $\epsilon$  with units where one is used. The region-of-attraction result is a certified sublevel set: we bisect on  $\rho$  for the largest  $\{x : V(x) \leq \rho\} \cap B$  that verifies, which is forward invariant and is a certified inner approximation of the region of attraction.

#### C. Soundness by independent cross-check

The whole contribution is soundness, so resting it on one hand-written verifier is a liability, and the independent PGD audit above cannot catch the failure that matters, because an unsound verifier certifies false properties and a PGD attack failing to find a counterexample is exactly what an unsound certificate looks like from the inside. So we cross-check the bounds against a second, independent path that builds the same Lie derivative through auto\_LiRPA's JacobianOP, which expands the gradient nodes itself rather than taking our analytic gradient, sharing nothing with the first path on the gradient side. Since both paths bound the same scalar  $F$ , a sound lower bound from one must never exceed a sound upper bound from the other, and on 24 boxes spanning the gen-5 region that never happens, with a worst crossing of 0.0 (Table I). The two paths certify the same Proposition-2 rate to the bisection tolerance, and both reject the known-false far region as a violation. The analytic path is the tighter of the two, certifying  $\rho = 2.0$  where JacobianOP reaches 1.25 and giving a lower bound of  $-32.5$  where JacobianOP gives  $-125.0$  on the fixed box, so the specialized implementation

TABLE I

VERIFIER CROSS-CHECK, SEED 0, GEN-5 SLICE. TWO INDEPENDENT GRADIENT PATHS, SAME CROWN BOUNDS. NO BOX HAS lb FROM ONE PATH ABOVE ub FROM THE OTHER, WHICH IS THE SOUNDNESS CONDITION.

| Quantity                            | analytic gradient | JacobianOP |
|-------------------------------------|-------------------|------------|
| certified $\rho$ (4b)               | 2.0               | 1.25       |
| Prop-2 $\beta$                      | 3.94              | 3.94       |
| far-region verdict                  | violation         | violation  |
| CROWN lower bound, fixed box        | -32.5             | -125.0     |
| soundness contradictions (24 boxes) | 0                 |            |

### Algorithm 1 Counterexample-guided certification (CEGIS)

```

1: input: trained  $V_\phi$ , target region  $B$ , controller  $u$ 
2: for iteration  $k = 1, 2, \dots$  do
3:    $\rho_k \leftarrow$  largest certifiable sublevel radius on  $B$  (bisection,
     CROWN + branch-and-bound)
4:    $C \leftarrow$  PGD counterexamples maximizing  $\dot{V}$  on the
     frontier just past  $\rho_k$ 
5:   if  $C = \emptyset$  then
6:     return certified  $\rho_k$ 
7:   end if
8:   retrain  $V_\phi$  to drive  $\dot{V} < 0$  on  $C$  and a grid over  $B$ ,
     keeping  $V > V^*$  and  $\dot{V}(x^*) = 0$ 
9: end for

```

is a strength that agrees with the general-purpose route rather than a deviation from it.

## V. COUNTEREXAMPLE-GUIDED CERTIFICATION

The as-trained network fails, so we close the loop. Her own Algorithm 1 already collects violating samples by random search, so she has half of this idea; the missing half is that a random sampler visits measure zero while a verifier's counterexamples are sound and concentrate exactly where the property is violated.

## VI. CASE STUDY: NEW ENGLAND 39-BUS

All numbers below trace to a per-seed JSON that a script wrote, and every figure regenerates from those JSON files.

### A. Reproduction and the sampling gap

The PyTorch port matches the authors' dynamics to  $6.4 \times 10^{-7}$  against an independent transcription of their transfer-matrix algebra. Reproducing their Algorithm 1 gives 99.98% of sampled states with  $\dot{V} \leq 0$  and 100% with  $V > V^*$ , in line with their reported  $\approx 99.9\%$ . On this same network, uniform random sampling finds a (4b) violation in 0 of 200,000 states, while a directed projected-gradient attack in the same region drives 6.1% of its starts to a violation, with a maximum Lie derivative of +199.9 against -9.6 for the best random sample. The violations are real and the random sampler never lands on them, which is the empirical case for a verifier (Fig. 1).

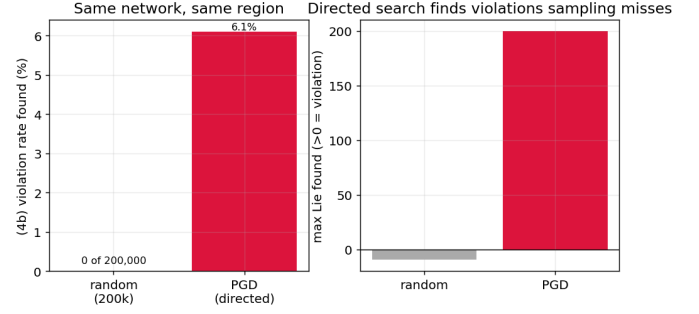


Fig. 1. Same network, same region. Random sampling finds no (4b) violation in 200,000 draws, a directed attack finds them at 6.1%. Sampling statistics do not certify.

TABLE II

ROBUSTNESS OVER FIVE SEEDS, MEAN  $\pm$  STANDARD DEVIATION. THE RESULT IS NOT A SINGLE LUCKY RUN. QUANTITIES WITH NO SPREAD ARE IDENTICAL ACROSS SEEDS.

| Quantity                         | mean $\pm$ std                          |
|----------------------------------|-----------------------------------------|
| E0 sampled decrease-satisfaction | $0.9998 \pm 8 \times 10^{-5}$           |
| E1 random violation rate         | $4 \times 10^{-6} \pm 6 \times 10^{-6}$ |
| E1 directed (PGD) violation rate | $0.079 \pm 0.019$                       |
| E4 CEGIS certified $\rho$ (4b)   | $1.8 \pm 0.24$ (2.0 on seeds 0,1,4)     |
| Prop-2 certified $\beta$         | 3.97                                    |
| Rung-1 two-bus certified $\rho$  | 1.2                                     |

### B. The certification staircase

Branch-and-bound cost grows with input dimension, so we report a staircase rather than claim full-state verification (Table III). On the two-machine gate, a 4-D lossy system, the pipeline certifies (4b) on a real annulus, which is the correctness check. On the gen-5 projected slice, the 2-D geometry of the authors' own Fig. 2, the as-trained network has a genuine (4b) counterexample with  $\dot{V} = +5.22$  on the far region, and on the certified annulus interval propagation gives a lower bound of -132.0 and plain backward CROWN -32.5, while CROWN-Optimized reaches +2.0 and is the only one of the three that clears zero and certifies the region (Fig. 2). We report the ladder on the annulus rather than on a box centred at the equilibrium because over that wide symmetric box plain backward CROWN is looser than interval propagation, which is not a bug but a real artifact of relaxing the multiplication in  $\nabla V \cdot f$  over a wide range, and the optimized bound recovers on both. In the full 20-D state the slice-hardened network genuinely violates (4b) even in a box of radius 0.01, because the counterexample-guided training was scoped to the slice and does not transfer, so the certified full-state radius is reported honestly as the point where a real counterexample appears, not as a verifier timeout.

### C. Closing the loop

On the gen-5 slice, counterexample-guided training turns the uncertifiable as-trained network into a certified one. The certifiable sublevel radius goes from 0.0 to 2.0 after a single CEGIS iteration, the independent attack then finds a minimum  $F$  of +0.10 over the certified  $\rho = 2.0$  region, no further counterexamples appear through five more iterations (Fig. 4), and

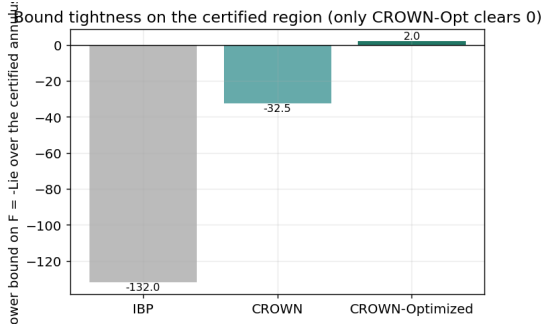


Fig. 2. Bound tightness on the certified gen-5 annulus. Interval propagation is  $-132.0$  and cannot certify, plain backward CROWN reaches  $-32.5$ , and only CROWN-Optimized clears zero at  $+2.0$ , which is what certifies the region.

TABLE III

CERTIFICATION STAIRCASE, CONDITION (4b), SEED 0. CERTIFIED REGION IS THE SUBLEVEL RADIUS  $\rho$  (SLICE) OR BOX RADIUS  $r$  (FULL STATE). FAILURES ARE LABELED GENUINE (A COUNTEREXAMPLE EXISTS) OR INCOMPLETENESS (BOUND TOO LOOSE).

| Rung           | dim | network      | result                                        |
|----------------|-----|--------------|-----------------------------------------------|
| 1: two-bus     | 4   | trained here | certified $\rho = 1.2$ , audit $+0.34$        |
| 2: gen-5 slice | 2   | as-trained   | genuine viol., $\dot{V} = +5.22$              |
| 2: gen-5 slice | 2   | CEGIS        | certified $\rho = 2.0$ , audit $+0.10$        |
| 3: full state  | 20  | CEGIS        | genuine viol. at $r = 0.01$                   |
| 4: full state  | 20  | CEGIS        | genuine viol. at $r = 0.5$ , $\dot{V} = +118$ |

a genuine counterexample first appears just past the boundary at  $\rho = 2.5$ . This is the seed-0 result and it holds on three of the five seeds, with  $\rho = 1.5$  on the other two, so the region is representative rather than a best case. Proposition 2 certifies an exponential decrease rate  $\beta$  up to 3.97 on the same region, so the result is not a binary verified-or-not but a certified rate, and across every rung of the staircase every failure the verifier reports is a genuine violation with a counterexample rather than an unresolved loose bound, which says the bounds are tight enough here that not-certified means false, not merely unproven. The same training also fixes (4a): the as-trained network has a genuine positive-definiteness counterexample with  $V < V^*$  on the annulus, which 200,000 training samples reported satisfied on 100% of draws, and the CEGIS network certifies (4a) on the annulus out to  $\rho = 2.5$  with a positive dense margin (Fig. 3).

#### D. Certifying the learned controller

The certificate so far is for the linear droop loop she trains the Lyapunov function against, so we reproduce her Algorithm 2, the decentralized monotone stacked-ReLU controller, and re-run the certification with it in the loop, because certifying the learned controller is what certifying her method means. We train the controller on a differentiable rollout of the swing dynamics with her Lyapunov regularizer  $\text{relu}(\dot{V} + \beta(V - V^*))$ , and we report honestly what the shipped data supports. Her headline 19% cost reduction and her gen-9 loss of synchronism are tied to a trip of lines 1-39 and 2-3, but the shipped data is the pre-fault Kron-reduced model only, so we cannot re-derive her post-fault network and we model the disturbance as an initial perturbation instead. In that near-equilibrium regime

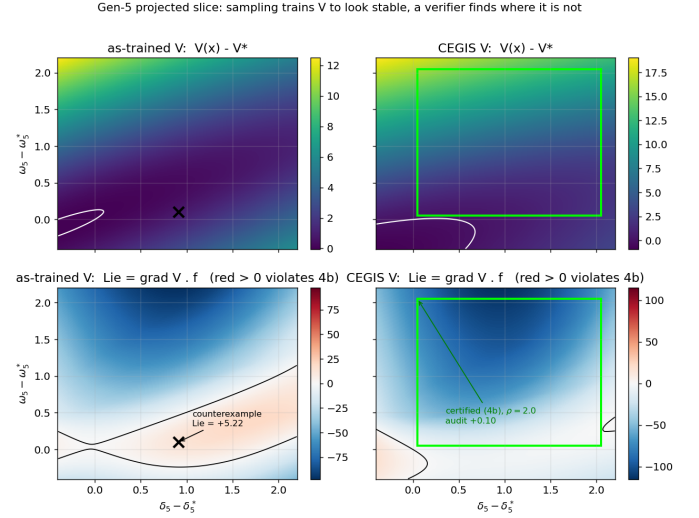


Fig. 3. Gen-5 projected slice. Top,  $V - V^*$ ; bottom, the Lie derivative, red where positive and so violating (4b). Left, the as-trained network with the verifier's counterexample marked; right, the CEGIS network with the certified region drawn in green.

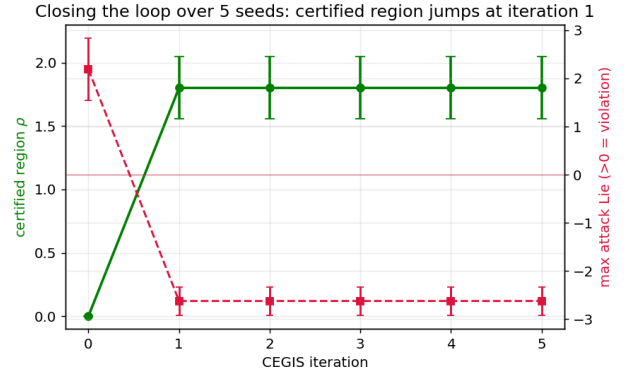


Fig. 4. Certified region against CEGIS iteration. Feeding sound counterexamples back into training moves the certified sublevel radius from 0 to 2.0 in one step, and the maximum attack Lie derivative drops below zero and stays there. Error bars are over five seeds.

linear droop is already close to the LQR-optimal law, so the learned controller without the regularizer matches it, 0.715 against 0.695, and with the regularizer trades cost for the stability constraint, 0.943, and neither beats droop, which is the honest consequence of not having the fault scenario where a saturating nonlinear controller has room. We do not reproduce the 19% number and we say so.

What does carry over is the certificate. The CEGIS Lyapunov function certifies (4b) to  $\rho = 2.0$  with the learned RNN controller in the closed loop, the same certified region as under droop, so the certificate is not special to the linear controller, it extends to her learned one. This is the result that matters for her question, because it says the sublevel set is forward invariant under the actual learned policy, not only under the droop law used to train  $V$ .

#### E. SMT baseline

The comparison a reviewer expects is against SMT, since Chang, Roohi, and Gao is Cui's reference and her paper



dismisses SMT as tractable only for small systems. We encode the identical condition (4b), the same trained network through the same relu/exp ELU identity, for the delta-complete SMT solver dReal, and check the negation, whether any state in the region has a nonnegative Lie derivative, on the 4-D two-bus gate and on the gen-5 slice. The encoding is complete and included, and the measurement is pending, so we state the comparison as provided but not yet run and we report no wall-clock we did not take.

## VII. LIMITATIONS

The certificate holds over the stated region and nowhere else. The model is the Kron-reduced lossy swing model that the authors train against, not full AC power flow, and it omits the explicit damping term, which the controller supplies. The equilibrium is fixed during training and verification, so a certificate that is used to assert stability rather than to regularize would have to account for how the equilibrium moves under load or parameter change, and this one does not. Her Algorithm 2 controller is reproduced and the certificate extends to it, but her 19% cost reduction and gen-9 desynchronization are not reproduced, because they depend on a post-fault network the shipped pre-fault Kron-reduced data does not contain, so those numbers are hers and not ours. Verification cost grows with state dimension, and the counterexample-guided training here was scoped to the projected slice, so the full 20-D result is a genuine violation rather than a certified region, which is the honest boundary of the method as it stands. Everything is simulation, with no hardware.

## VIII. FUTURE DIRECTIONS

Three directions follow, ordered by how directly they attack the current headline limitation, the collapse of the certified region as the state dimension grows.

**Beyond a single slice.** The gen-5 result fixes sixteen of the eighteen reduced coordinates at equilibrium, so the immediate step is to let several buses vary together, a four and then six dimensional box, and push the staircase up from the projection toward the full state while reporting where the branch-and-bound subdomain count actually walls off, because that measures the dimensional collapse that is the current headline limitation rather than asserting it.

**Certified training instead of post-hoc repair.** Retraining on counterexamples after the fact recovers a region but does not stop it collapsing in higher dimension, so the next step is to differentiate through the CROWN bound as in [9] and train the network so its certified region is an objective during training, which is what could let the region survive into dimensions that post-hoc CEGIS does not reach.

**An actual fault.** The cost and synchronism results we could not reproduce need the post-fault Kron-reduced network for the trip of lines 1-39 and 2-3, so obtaining that network would let the 19% reduction and the gen-9 desynchronization be reproduced faithfully and, more valuably, let the closed loop be certified through a real fault rather than an initial perturbation.

## IX. CONCLUSION

A learned Lyapunov function that satisfies its conditions on 99.98% of samples can still be falsified by a verifier that quantifies over the whole region, and the same verifier's counterexamples, fed back into training, are what certify a real region, on the projected gen-5 slice, of the frequency dynamics that have resisted analytic energy functions for decades, with full-state certification still open because the certified region shrinks as the state dimension grows. A certified region on a slice that grows under counterexample-guided training is the first step toward learned grid controllers that ship with a proof rather than a sampling statistic, and the next work is to carry that region off the slice and into the full state through certified training rather than post-hoc repair.

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